The vibes that I summoned…

Thoughts on working with AI agents.

Chris Pahl · 2026

The Spectrum (on Reddit)

Full manual
every keystroke yours

Full vibecode
just make it work

⟵ more control · more understanding | more productivity · more delegation ⟶

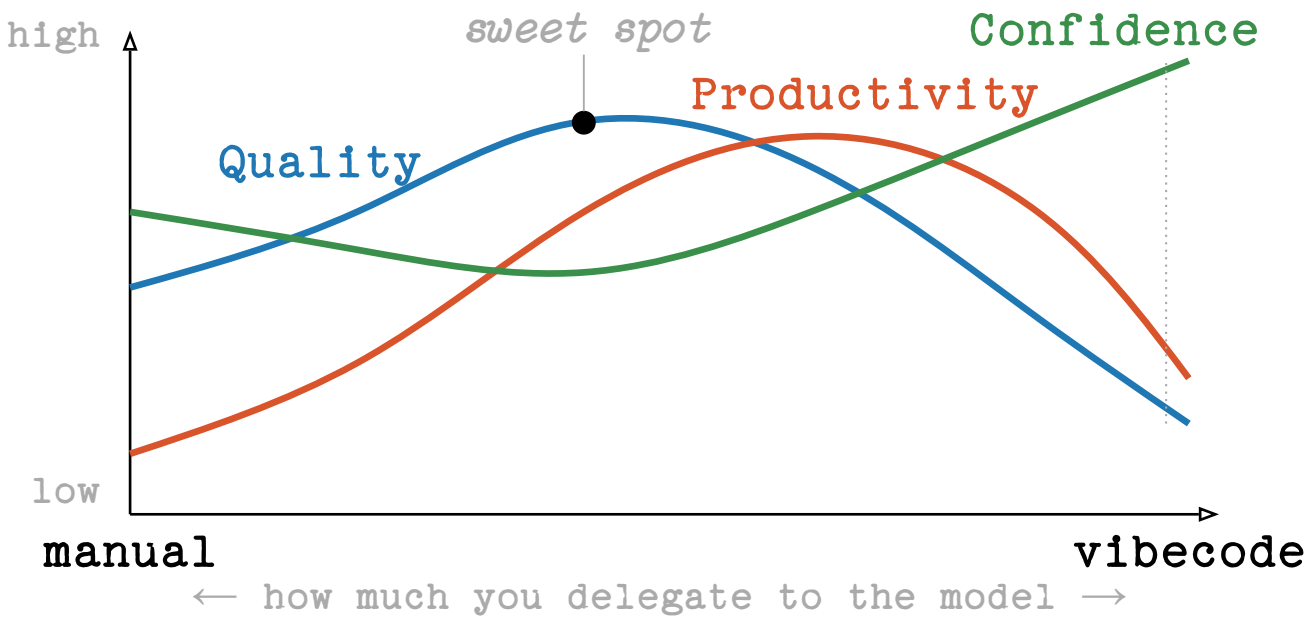
General tendency:

Using GenAI is a tradeoff between control and productivity.

Prompt:

“Imagine Donald Duck as realistic human. Remove the hat.”

My take: Quality vs Productivity



Quality peaks in the middle. Productivity follows it down. Confidence diverges from both -> Dunning-Kruger zone!

16%

But there's already data:

-19%

slower with AI

experienced devs · own mature repos · predicted +24%

METR · RCT · 2025

~40%

insecure programs

Copilot output across 89 security-relevant scenarios

NYU CCS · 2021

8×

more duplicated code

block-level clones up · refactoring 24% → 10%

GitClear · 211M LoC · 2024

-7.2%

delivery stability

drop with AI adoption · 39% distrust AI code

DORA / Google · 2024

29.5%

Python with CWEs

AI-generated code in real GitHub repos · 43 CWE categories

Fu et al. · ACM TOSEM · 2025

↓

trust = less scrutiny

more trust in GenAI ⟹ less critical thinking applied

Lee et al. · Microsoft + CMU · CHI 2025

Good use cases

rubber-ducking ideation

explain unfamiliar code test generation

pair programming boilerplate refactoring assist

learning accelerator documentation drafts naming things

summarisation regex / SQL crafting edge-case brainstorming

translation code review companion mock data / fixtures

error decoding search engine log analysis proofreading

automation

Risks for all

ecological cost

Convincing hallucinations

schools

licensing

Atrophy

concentration

misinformation at scale

Prompt injection & MCP

content degradation

No juniors hired

hardware crisis

operator bias

recursive collapse

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communities thinning out

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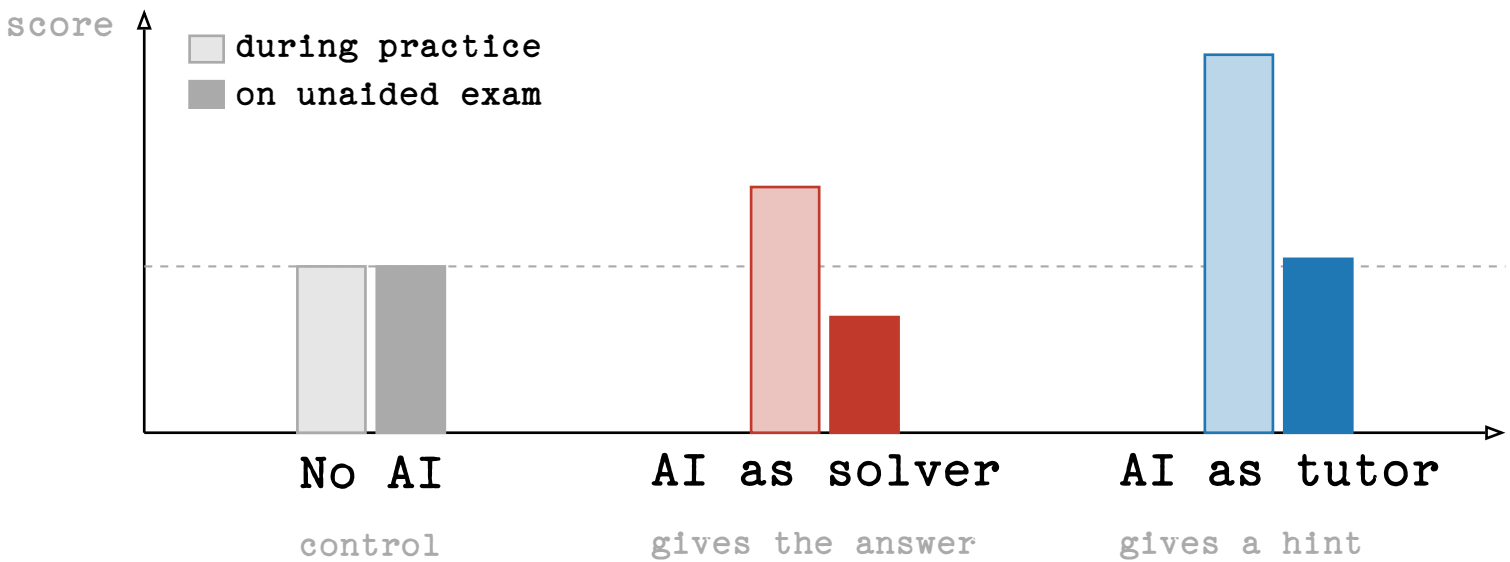
Exhibit A: schools.

Maybe it's not so much about the tool —
but about how we use it?

Same tool, opposite outcomes



Bastani et al., *Generative AI Can Harm Learning*, SSRN 2024

Are *you* team tutor or team solver?

Keeping up with Claude

Our team's Claude Code account · April 2026

98.7%

suggestions accepted as-is

1.3% rejected. Is that a review process?

27,757

lines accepted last month

≈ 925 LoC/day · careful review ≈ 100–200 LoC/hour

Reviewing 27,757 lines carefully ≈ 138 h.

Human cognitive biases

Automation bias

We accept machine suggestions more readily than human ones.

click accept · review later · maybe

Anchoring

The first suggestion shapes the solution space — even when it's wrong.

the model frames the problem before you do

Confirmation bias

We hear what we already believe.

the prompt contains the answer we want

Authority bias

Eloquent + fast + confident reads as expert.

fluency ≠ correctness

Dunning–Kruger

We mistake competence we observe for competence we have.

"this is what I would have written"

Illusion of explanatory depth

We think we understand — until asked to explain.

"yes, I read the diff" → bug

Statistical model biases

Sycophancy

It's easy to talk the model out of a correct answer.

echo chambers — but for code

Historical / representation bias

Trained on the web — heavily modern, English, Western.

defaults track what's common, not what's right

Attention bias

First and last things dominate; the middle evaporates.

rules buried in long context get ignored

Omission bias

Rare and novel answers get suppressed by common ones.

the model is bad at being weird

Hen & Egg

1. If you don’t know what good software looks like —
   how do you write the right prompt?

2. If you can’t understand what the model just generated —
   how do you verify it?

3. If you can’t code (anymore) —
   how can you understand the diffs?

4. If you do not know what you build inside-out —
   how do you learn new designs?

How do *you* keep up with Claude?

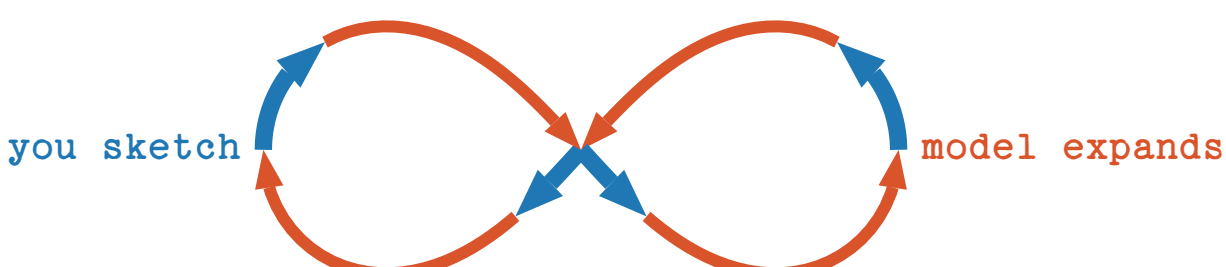
Five habits that helped me

1. Sketch first, generate then.
2. Split the work - keep some of it manual.
3. Have a real verification strategy.
4. Don’t make yourself replaceable.
5. Treat the AI like a colleague.

1. Sketch first, generate then

You set the anchor, you stay in the loop.

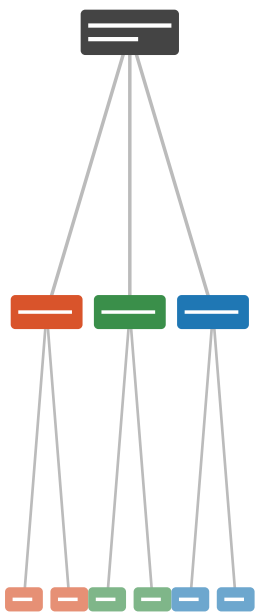
- Sketch the SQL query yourself,
  Then ask the model to review and optimise.
- Write the function signature and the docstring.
  Then let the model fill the body.
- Put TODO comments in your code
  Then let Claude work on them.



2. Split the work, do some manual

The parts you write are the parts you understand.

- Claude does not perform well, if given too big scopes.
  Split tasks up and prompt them individually.
- Claude can help you split up work.
  It just tends to work on the task right away.
- It is easy to lose understanding without noticing.
  Keep doing important things manually!
- Large diffs make it easy to get lost.
  Commit often so diffs stay reviewable.



3. Verify strategy before code

There are no shortcuts to quality.

Before you accept a generated change

name the signal that would tell you it’s wrong.

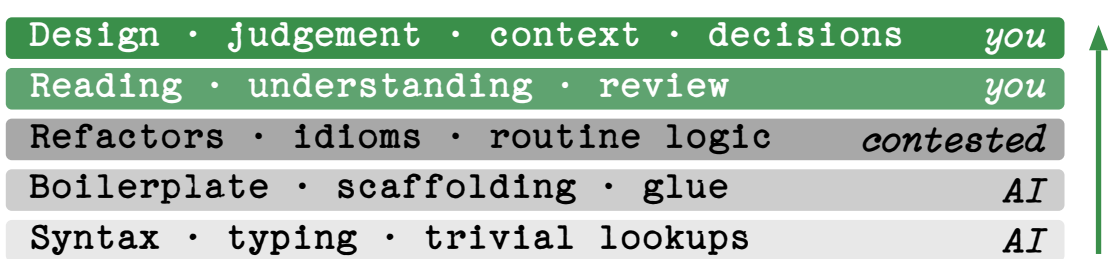
- Comparison against a reference implementation
- Property-based tests, fuzzing, …
- Type checkers, linters, static analysers
- **/review** and manual review by colleagues
- Replay against real production data
- Work actively on the code

Example: Noise library for Dart — I don’t speak Dart.

4. Don’t make yourself replaceable.

You only get replaced if you make yourself replaceable.

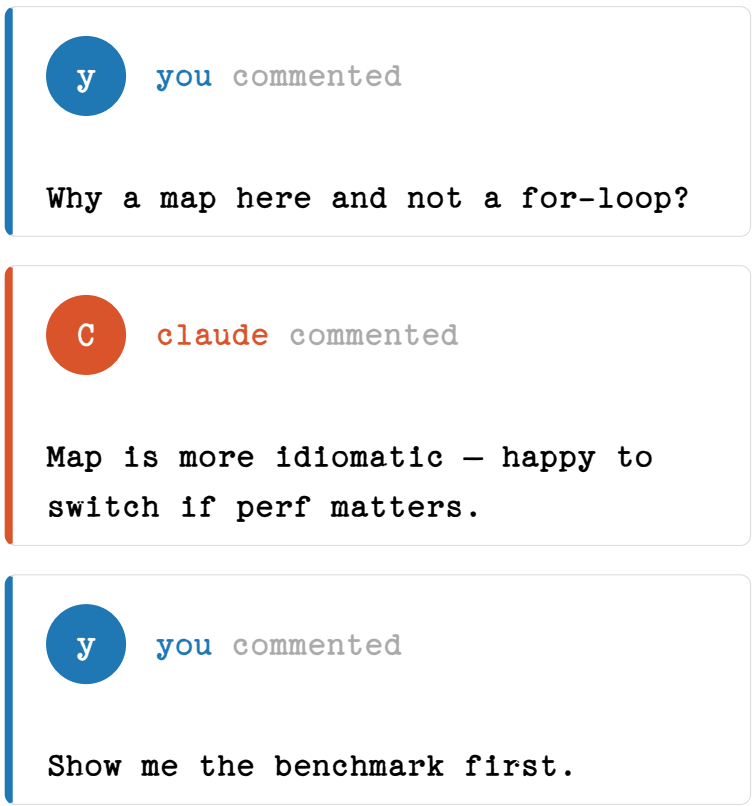
- Code was the source of truth before, that’s changing.
  Now it is context given via design documents.
- Spend the time Claude saves you on the parts it’s bad at.
  Namely: decisions, judgement, context, design.
- Reading code is much more important now than writing code.
  Keep your tools sharp. Be able to work without Claude.



5. Treat Claude like a colleague

Talk to it like your seat neighbor.

- **Explain the tasks at hand like you would to a junior colleague.**
  A very eager, junior colleague with seemingly infinite capacity.
- **Push back. Ask for alternatives. Let it explain. Disagree.**
  If you'd reject a colleague's PR for that reasoning, reject the model's.



Bonus: Programming as Theory Building

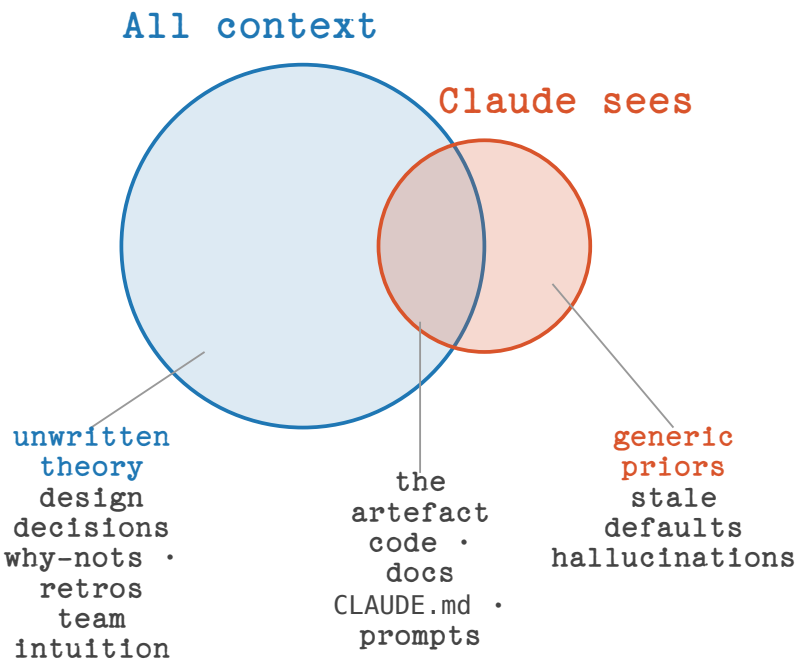
Peter Naur, 1985

- The code is a by-product. What you're really building is the theory — your team's shared mental model of the problem.
- Documentation captures the artefact, not the theory. The bugs, the dead ends, the "why not this?" decisions live in people's heads.
- This is why handovers fail. The theory doesn't transfer cleanly.

Peter Naur, 1928–2016
Photo: Wikimedia Commons

Bonus: What that means for Claude

- The model has no theory — it sees the artefact and fills the rest in.
  Confidently. It won't tell you which is which.
- Bad prompt: "Write tests for this file"
  It freezes today's behaviour, bugs included. Again, no mention.
- Your job is still to build the theory.
  The AI helps you write the code that follows from it.



Questions or Disagreements?

Homework:

PROGRAMMING AS THEORY BUILDING.